Credit Risk Classification Using Logistic Regression

Executive Summary & Predictive Performance Evaluation

Dataset: South German Credit (UCI, N = 1,000) **Methodology:** Logistic Regression & Validation Threshold Optimization

EXECUTIVE OVERVIEW

This project constructs an interpretable baseline credit-scoring model using logistic regression to classify applicants according to the recorded risk label in the South German Credit dataset (700 good risk, 300 bad risk). The study integrates exploratory data analysis, stratified train/validation/test evaluation, decision threshold selection, and formal odds-ratio inference on borrower risk factors.

KEY EMPIRICAL FINDINGS & STATISTICAL INFERENCE

- **Loan Duration Risk Association:** Loan duration exhibited the most statistically strong relationship with bad credit risk. Each additional month increases the odds of bad-risk classification by approximately **3.7%** (Odds Ratio = 1.037, $p < 0.001$), holding other borrower attributes constant.
- **Credit History & Savings Status:** Categories indicating critical credit history and higher recorded savings status showed statistically significant protective effects, significantly reducing default odds compared to reference categories ($p < 0.05$).
- **Non-Significant Predictors:** Requested credit amount ($p = 0.278$) and applicant age ($p = 0.407$) were not statistically significant at the 5% level conditional on duration, history, and savings.
- **Model Fit:** Estimation on the 800-observation training sample yielded a McFadden Pseudo-R^2 of 0.1427 (Log-Likelihood = -418.97).

HELD-OUT TEST SET PERFORMANCE COMPARISON (N = 200)

To address class imbalance, a probability decision threshold of **0.34** was chosen on validation data using a minimum 50% bad-risk recall rule before evaluating the held-out test set:

Performance Metric	Baseline Model (Threshold 0.50)	Final Model (Threshold 0.34)	Operational Impact
Accuracy	73.0%	**67.0%**	Reflects higher false positive flags at lower threshold
Bad-Risk Precision	60.7%	**45.6%**	Share of flagged applicants who are actually bad risk
Bad-Risk Recall	28.3%	**51.7%**	**+82.7% increase** in detecting actual bad-risk applicants
F1-Score	38.6%	**48.4%**	Improved harmonic mean of precision and recall
ROC-AUC	0.698	**0.698**	Moderate class ranking discrimination across thresholds

> **Conclusion & Scope Note:** Lowering the threshold to 0.34 meaningfully improves bad-risk detection (recall increases from 28.3% to 51.7%) at the expense of precision. This model serves as an educational baseline demonstrating trade-offs in credit risk decision-making rather than a production-ready system.